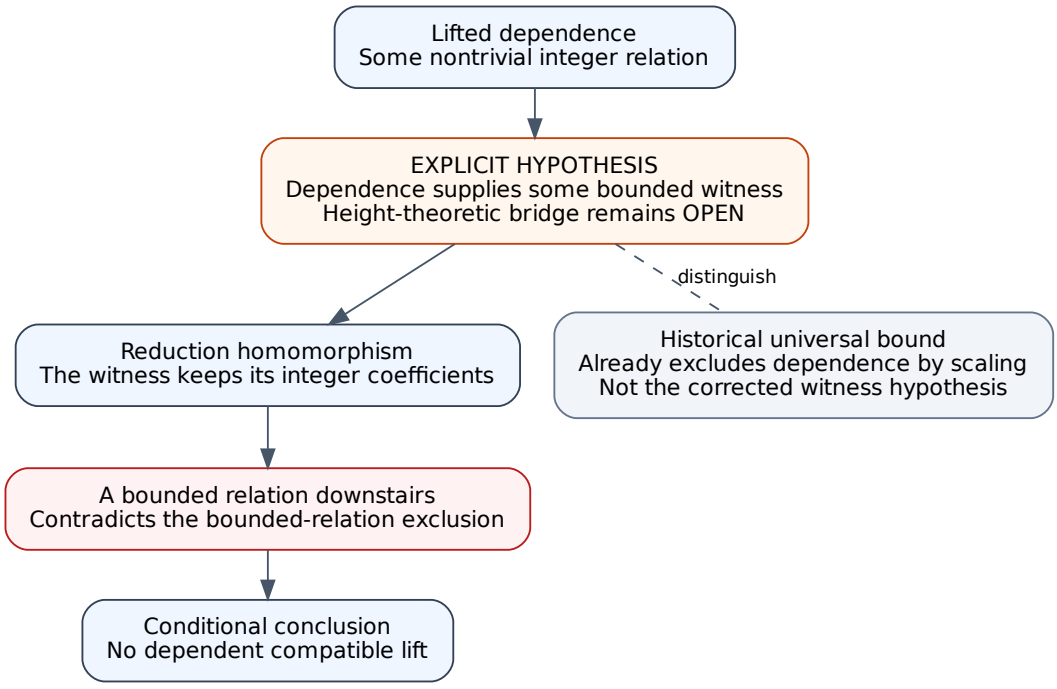
Corrected bounded-witness implication

Xedni proof correction | 30 October 2026

Exact issue and correction

The historical statement bounds every nontrivial relation by a fixed B. Scaling any nonzero relation violates that bound, so the hypothesis already implies independence. The corrected hypothesis says that if a relation exists, some nontrivial bounded witness exists. The witness can differ from the original relation.

Conditional implication and open bridge



Orange identifies an explicit mathematical hypothesis, not a proved height bridge. Native verification of the new five-theorem source is pending. The sealed original build is not validation of this successor.

Written argument and proof map

Let rho: H -> G be an additive homomorphism and let a lifted tuple reduce to the fixed tuple downstairs. Suppose a nontrivial lifted relation exists. The bounded-witness hypothesis supplies another nontrivial lifted relation with every coefficient at most B in absolute value. Reduction preserves its coefficients, giving a forbidden bounded relation downstairs. This proves the conditional exclusion. For all compatible lifts, the hypothesis is supplied for each tuple using the same B; the chosen witness may vary.

Controls against the original circular hypothesis

The dependent zero tuple has a nontrivial witness bounded by one. The same tuple has nontrivial relations beyond every fixed bound. These two controls distinguish existence of a bounded witness from a universal bound on all relations.

Declaration	Current status
relation_reduces	Implemented; native check pending
no_dependent_lift	Implemented; native check pending
all_compatible_lifts_blocked	Implemented; native check pending
zero_tuple_has_bounded_witness	Implemented; native check pending
zero_tuple_has_unbounded_relations	Implemented; native check pending

Evidence and remaining obligations

Source: formal/WitnessObstruction.lean. Toolchain: leanprover/lean4:v4.35.0-rc4. Mathlib: 021c46bdd042f67b538b9f9871d556a93713a9a9. All five declarations request a foundation audit. Dependency hashes bind inputs and do not attest a completed build. The original sealed bundle remains byte-exact.

Primary source: Jacobson et al., Analysis of the Xedni Calculus Attack, sections 3-4, Lemma 4.1; https://pages.cpsc.ucalgary.ca/~jacobs/PDF/xedni.pdf. The lemma supplies a witness under height/discriminant and Lang assumptions; that height bridge and the actual curve-selection distribution remain open. No new experiment, general impossibility or performance claim is made.

Canonical graph review: this proof correction changes no curve, measured row, atlas status or performance edge. Editable diagram: implication.dot; full written report: RESULT.md.